

PYP

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- i. Draw the timing diagram for the UART interface to transmit a 7-bit ASCII character of value **0x1B**.

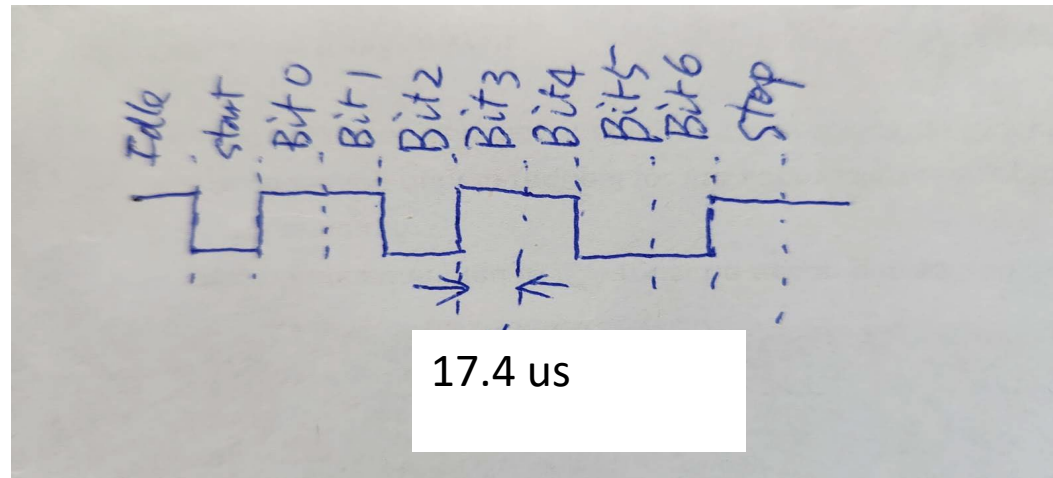
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57,600 bps $\rightarrow$ time interval = $1/57,600$ sec = 17.4×10^{-6} sec = 17.4 μ s (1 mark)

0x1B $\rightarrow$ 001 1011 $\rightarrow$ LSB first $\rightarrow$ 1101 100 $\rightarrow$ idle 1 start 0 stop 1: (1)0 1101 100 1



(1 marks)

(3 marks)

(CONT'D)

A UART interface has been configured as: 7 data bits, 1 stop bit, no parity check, and baud rate of 57,600bps.

ii. Derive the maximum data transmission rate of the UART interface (in bits per second).

No. of bits/frame = $1 + 7 + 1 = 9$ (bits) (1 mark)

Max no. of frames/sec = $57,600 / 9 \rightarrow 6,400$ (1 mark)

Max data transmission rate $\rightarrow 6,400 * 7 \rightarrow 44,800$ (bps) (3 marks)

(CONT'D)

A UART interface has been configured as: 7 data bits, 1 stop bit, no parity check, and baud rate of 57,600bps.

iii. What is the minimum clock frequency for a SPI interface to transmit data faster than the UART interface using the given configurations described above?

SPI: 1 data bit per clock (no syn bits, etc.) → Data rate=clock frequency (2 marks)

SPI clock frequency > 44,800 Hz (3 marks)

1. (a) The following points were raised during a product design review meeting. State whether you **agree or disagree** with the statement and give a **very short explanation** to justify your answer. Target product is an ultra-thin lap-top computer with 32GByte System Memory and 512GByte Storage Memory.

(i) NOR Flash is a good choice for the System Memory as it supports Execute-In-Place.

(5 marks)

(ii) HDD is a better choice than SSD for Storage Memory because it has an almost infinite erasure cycle limit.

(5 marks)

(iii) DRAM is more suitable than NAND Flash for use as the storage memory because it allows infinite number of read/write operations.

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(iii) DRAM is more suitable than NAND Flash for use as the storage memory because it allows infinite number of read/write operations.

(i) ~~Agree~~ / Disagree

(5 marks)

- NOR Flash has finite program-erase cycles.
- System memory on a Laptop will have a lot of read-write operations and so DRAM will be more suitable.

(ii) ~~Agree~~ / Disagree

- The lap top needs to have an ultra-thin form factor; HDD size is much larger and thicker than SSD
- The lap top is with a lot movement; HDD is prone to crashing

(iii) ~~Agree~~ / Disagree

- Storage Memory needs to preserve the information during power shutdown.
- DRAM is a volatile memory and information will be lost once power supply is cut.